

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2022-2023
FULL MARKS: 75

CSE 4617: Artificial Intelligence

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) In the movie, Colossus – The Forbin Project, a US defense computer tasked with ensuring peace and security encounters a Soviet computer with the same goal. They exchange information and decide that the goal is best achieved by jointly controlling all nuclear weapons and using threats of destruction to force humans to drop all aggressive war plans.
- In another movie, Transcendence, a professor of Artificial Intelligence (AI) is gunned down by anti-AI terrorists. Before he dies, his brain is uploaded to a supercomputer that was designed to integrate all human emotions with the accumulated knowledge of humanity. The machine becomes conscious and begins to quickly outrun the human race, threatening to take over the world.
- In light of the recent developments in the field of Artificial General Intelligence, provide your recommendations on how these situations mentioned above can be avoided. Your discussion should identify the common cause behind the problematic situations, what we should do when developing the agents, and what the agents should do.

7
(CO3)
(PO3)

Solution:

In both cases mentioned in the given scenario, the risk comes from the possibility that the machine's conscious goals differ from those of its human progenitor due to poorly defined objective.

It is impossible to anticipate all the ways in which a machine pursuing a fixed objective might misbehave. There is good reason, then, to think that the standard model is inadequate. We do not want machines that are intelligent in the sense of pursuing their objectives; we want them to pursue our objectives. If we cannot transfer those objectives perfectly to the machine, then we need a new formulation in which the machine is pursuing our objectives, but is necessarily uncertain as to what they are.

When a machine knows that it does not know the complete objective, it has an incentive to act cautiously, to ask permission, to learn more about our preferences through observation, and to defer to human control. Ultimately, we want agents that are provably beneficial to humans.

Rubric:

- 2 points for identifying the reason for the problems
- 3 points for mentioning what we should do
- 2 points for describing what the agent should do

- b) Consider that we formulate a scenario as an AI problem where the performance measure is concerned with just the first T time steps of the environment and ignores everything thereafter. Show how that affects the decision made by a rational agent using example(s).

5
(CO2)
(PO2)

Solution:

Suppose that for a particular problem, in any state there are two action choices, a and b , and consider two cases: the agent is in the state S at time T or at time $T - 1$. Let, from state S , the action a reaches state S' with reward 0, while action b reaches state S again with reward 1. From the state S' , the action a/b gains a reward of 10. At time $T - 1$, it is rational for the agent to take action a with expected total reward 10 before the time is up. But at time T , it is rational to take action b with expected total reward 1, since 10 cannot be obtained before the time is up.

(Other real-life examples demonstrating the same phenomenon will work)

Rubric:

- 5 points correct formulation demonstrating the effect

2. Consider the problem of moving k ($k \geq 1$) knights from k starting squares $s_1, \dots, s_k$ to k goal squares $g_1, \dots, g_k$ on a chessboard with infinite size. Here, no two knights can occupy the same square at the same time. The goal is to transfer all the knights using the fewest possible actions.

The scenario can be formulated as a search problem in two ways:

- F_1 : An action involves moving at most k knights simultaneously at a time step.
- F_2 : An action involves moving only one knight at a time step.

Now, we consider designing a heuristic for the given scenario. Assume that h_i is an admissible heuristic for a relaxed problem where we move the i^{th} knight to goal g_i by itself, ignoring the other $(k - 1)$ knights. Also we have three proposed heuristics for the given scenario:

- $h_{\min} = \min\{h_1, \dots, h_k\}$
- $h_{\max} = \max\{h_1, \dots, h_k\}$
- $h_{\text{sum}} = \sum_{i=1}^k h_i$

For **each of the possible formulations** of the given scenario, answer the following questions:

- a) Evaluate the admissibility of each of the proposed heuristics.

2×12
(CO3)
(PO3)

Solution:

Considering F_1 :

If h_i provides us the exact value, $h_{\max}$ would be exact for the relaxed problem where knights can land on the same square. Since h_i is admissible and hence no larger than the exact values, $h_{\max}$ is admissible.

Since $h_{\min}$ is no larger than $h_{\max}$, $h_{\min}$ is admissible.

h_{sum} is not admissible. Consider the scenario where each knight is one move away from the goal. Here, we can move all the k knights simultaneously to their respective goals. The optimal cost would be 1 here, whereas h_{sum} will provide a value of k .

Considering F_2 :

Since the problem defined earlier is a relaxation of this new problem, both $h_{\min}$ and $h_{\max}$ will remain admissible.

This time, h_{sum} is admissible because it is exact for the relaxed problem where knights can land on the same square.

Rubric:

- 1 point for each correct decision
- 3 points for each justification

- b) Recommend the most suitable one among the three proposed heuristics with proper justification.

2 × 3
(CO3)
(PO3)

Solution:

Considering F_1 :

$h_{\max}$ dominates $h_{\min}$ so it must be as good or better, whereas h_{sum} is not admissible, so it is of no use here.

Considering F_2 :

Since h_{sum} dominates the rest and is now admissible, we should choose this one.

Rubric:

- 1 point for identifying the correct one
- 2 points for the justification

3. As shown in Figure 1, Sulawesi, an Indonesian island, is divided into six provinces: Gorontalo (G), West Sulawesi (WS), South Sulawesi (SS), Central Sulawesi (CS), Southeast Sulawesi (SES), and North Sulawesi (NS).

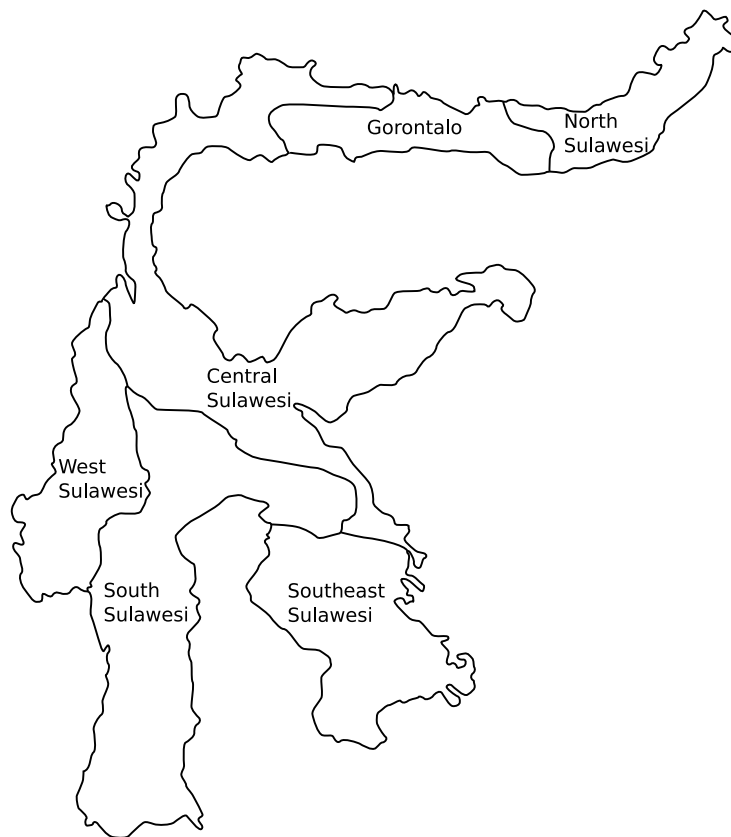


Figure 1: A Simplified Map of Sulawesi for Question 3

We want to color each province either red, green, or blue in such a way that no two neighboring regions (provinces that share their borders) have the same color.

- a) Formulate the scenario as a Constraint Satisfaction Problem (CSP) by identifying the variables, their possible values, and the constraints. Also, draw the constraint graph.

15 + 7
(CO2)
(PO2)

Solution:

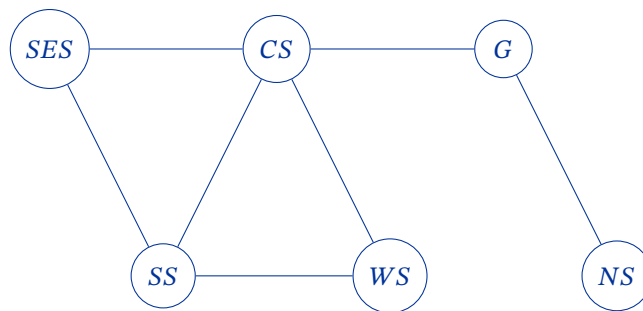
Variables: $\{G, WS, SS, CS, SES, NS\}$

Values: $\{red, green, blue\}$

Constraints:

- $G \neq NS$
- $G \neq CS$
- $WS \neq CS$
- $WS \neq SS$
- $SS \neq CS$
- $SS \neq SES$
- $SES \neq CS$

The constraint graph is as follows:



Rubric:

- 1 points for each variable
- 2 point for values
- 1 point for each constraints
- 0.5 points for each node
- 0.5 points for each correct edge
- 0.5 point for not including any wrong edge

b) Identify the possible cutset(s) of the constraint graph for converting it to a tree.

3
(CO1)
(PO1)

Solution:

CS and SS

Rubric:

- 1 point for each correct cutset
- 1 point for not including any wrong cutsets

4. Consider that people are given a choice between lotteries A and B and then between C and D , which have the following prizes:

$$A = [0.80, \$4000; 0.20, \$0]$$

$$C = [0.20, \$4000, 0.80, \$0]$$

$$B = [1.00, \$3000; 0.00, \$0]$$

$$D = [0.25, \$3000, 0.75, \$0]$$

Most people consistently prefer B over A , and C over D .

- a) Given $U(\$0) = 0$, show that the preferences exhibited by most people violate how a rational agent should act.

4
(CO1)
(PO1)

Solution:

We note that if

$$\begin{aligned} B &> A \\ \Rightarrow U(B) &> U(A) \\ \Rightarrow U(\$3000) &> 0.80U(\$4000) \end{aligned} \quad (1)$$

Then, if

$$\begin{aligned} C &> D \\ \Rightarrow 0.20U(\$4000) &> 0.25U(\$3000) \\ \Rightarrow 0.80U(\$4000) &> U(\$3000) \end{aligned} \quad (2)$$

Here, Equation 1 contradicts Equation 2. That means, given a rational agent prefers B over A , they should prefer D over C .

Rubric:

- 1 point for each final equation
- 2 point for conclusion

- b) Does the outcome in Question 4.a prove that humans are irrational? Justify your answer.

4
(CO3)
(PO3)

Solution:

No. One explanation for the apparently irrational preferences is the certainty effect: people are strongly attracted to gains that are certain. There are several reasons why this may be so:

- People may prefer to reduce their computational burden; by choosing certain outcomes, they do not have to compute with probabilities. But the effect persists even when the computations involved are very easy ones.
- People may distrust the legitimacy of the stated probabilities. I trust that a coin flip is roughly 50/50 if I have control over the coin and the flip, but I may distrust the result if the flip is done by someone with a vested interest in the outcome. In the presence of distrust, it might be better to go for the sure thing.
- People may be accounting for their emotional state as well as their financial state. People know they would experience regret if they gave up a certain reward (B) for an 80% chance at a higher reward and then lost.

In other words, if A is chosen, there is a 20% chance of getting no money and feeling like a complete idiot, which is worse than just getting no money. So perhaps people who choose B over A and C over D are not irrational; they are willing to give up \$200 of EMV to avoid a 20% chance of feeling like an idiot.

Rubric:

- 1 point for decision
- 3 points for reasoning